

Oluwatobi Melvyn Mayungbo

Data Analyst, Fraud and Risk

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SUMMARY

Data analyst focused on financial crime, fraud and risk. Built transaction fraud detection with per-attack analysis, a payment reconciliation system with audit trails, a compliance assistant over central bank regulation, and wallet screening against sanctioned and mixer addresses. Works in SQL, dbt and Python.

SKILLS

Payments and Risk: Transaction Fraud Detection, Regulatory Compliance (CBN/NDPA), Payment Reconciliation, Credit Risk Decisioning, Double-Entry Ledgers and Idempotency, Demand Forecasting, Web3 Fraud Risk Scoring

Data Analysis: SQL, Anomaly Detection, Pandas, NumPy

Engineering and MLOps: dbt-core, Python, Git, SQL, DuckDB, PostgreSQL, Redis, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, MLflow, Evidently, CI/CD (GitHub Actions), Pytest, Streamlit, React

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

PROJECTS

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.
- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.

Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.
- Engineered a multi-stage reconciliation pipeline using DuckDB and FastAPI, with temporal join logic and tolerance thresholds to handle asynchronous processor webhooks and bank settlement files.
- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Detects honeypot tokens, rug pull signs, hidden owner powers (mint, blacklist, pause), buy and sell tax, holder concentration, unlocked liquidity, and fund links to mixers, sanctioned wallets, exploiters and phishing addresses over 1 or 2 hops.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action